

Malayalam Palm Leaf Character Recognition using Gabor Features

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Abstract—The objective is to compare and interpret the decisions of the algorithm to recognise the palm leaf characters. Some of the machine learning models are been used like Logistic Regression, Support Vector Machine (SVM), Random Forest Classifier, Decision Tree, K-Nearest Neighbour (KNN), Multi Layer Perceptron (MLP), Navie Bayes, XGBoost, CAT Boost and ADA Boost are taken and compared to see the better results in recognizing the features. Hyperparameter is been used. And LIME is been implied in models for interpretability.

Index Terms—Palm leaf, Character recognition, Machine learning, k-means, Random forest.

I. INTRODUCTION

Palm leaf manuscripts represent an important cultural and historical heritage, with countless ancient texts and scriptures engraved on these delicate organic materials. Preserving and digitizing these manuscripts has become an important undertaking to protect this valuable source of knowledge for future generations. One of the main challenges of this effort is the accurate identification of characters from palm leaf manuscripts, which often exhibit irregularities in shape, size and discoloration due to the organic nature of the material. To address this challenge, researchers have developed innovative methods, feature-based character recognition, which combines the strengths of different feature extraction techniques to improve Accuracy and reliability of character recognition from palm leaf manuscripts. This combination of features from different sources and methods promises to overcome the challenges posed by palm leaf manuscripts, ensuring the preservation and accessibility of their rich textual content.

Character recognition based on palm leaf manuscript features leverages a diverse set of feature extraction techniques, such as geometric, textural, and structural features, to capture details complexity of each character.

Geometric features focus on the spatial arrangement of strokes and elements within characters, while texture features consider finer surface patterns and features. On the other hand, structural features analyze the relationship between different parts of a character. The next essential step is to train the recognition model. The kind of the recognition challenge and the hybrid characteristics in use determine which model should be used. This stage involves carefully labelling the data, giving each character a corresponding Malayalam symbols. By combining these feature sets, hybrid methods can handle the variability and complexity of characters on palm leaf manuscripts, handling variations due to ink fading, damage, and style. Handwritten writing thereby contributing significantly to preserve and disseminate the cultural and historical knowledge encoded in these ancient texts.

II. LITERATURE SURVEY

The relevance of optical character recognition and binarization has increased significantly in light of years of study. The dataset goes through preprocessing, such as noisy data removal and binarization, before using numerous methods. Certain techniques are used for feature extraction. The characters in photos of palm leaves were found using the contour approach in Reference [1]. The photos are converted to grayscale and then binarized to eliminate noise. CNN with many classes

was planned. In this study, 5-CNN layers are suggested. They achieved training accuracy of 98% and test accuracy of 96%. Reference [2] did not directly binarize the picture; instead, it employed division normalization. To reduce noise, the photos were afterwards converted to grayscale. They achieved 90% accuracy with the CNN model. The authors in [3] insights of broader recognition from palm leaves, offerings broader research about the above things more closely. Design the CNN and Construct CNN network and Compute weights and biases and train network and evaluate designed network for identified different classes. Reference [3] suggested three steps: (I) input picture enhancement by preprocessing; (II) segmenting the characters from a palm leaf; and (III) character recognition using a Custom Multilayer CNN. For custom CNN, the accuracy difference between training and testing is around 8%. View-based mechanisms were employed in Reference [4] to extract the characteristics from the characters on palm leaves. Artificial neural networks are used to classify extracted characters. Malayalam characters were categorized into forty-four classes with 98% accuracy. Reference [5] conducted study on a few Tamil letters. They have put out an RNN architecture. The data set was trained using the LSTM algorithm. To extract the features, they made use of OpenCV. They were able to get many Tamil characters with varying degrees of accuracy.

The authors in [6] presents a deep learning approach for recognizing cursive characters. In comparative analysis wavelet and curvelet and Gabor transform are carried out for face recognition and it is found that curvelet works better compared to others. Reference [7] presents about mathematical transform to extract the features from characters for recognition. The most complex deep model may take weeks to train using hundreds of machines equipped with expensive GPU and deep learning and machine learning is not easy for this classifiers. Reference [8] presents an effective approach for feature extraction and classification in Kannada handwritten character recognition. The method involves extracting suitable features, which include both structural features and global features obtained through wavelet transform. To derive structural features, the pre-processed image is divided into four quadrants, and various characteristics such as corner detection, aspect ratio, and quadrant density are computed for each of these zones. Reference [9] It takes a three-dimensional perspective, utilizing histogram computation and distance profile features to improve accuracy. By incorporating these techniques, the paper aims to enhance the recognition and preservation of characters within the intricate context of palm leaf manuscripts, contributing to the broader field of document analysis and cultural heritage preservation. Reference [10] this presents a feature extraction approach for offline identification of handwritten Malayalam characters based on extracting the count of adjacent ones in rows and columns. This method uses the horizontal and vertical count of adjacent pixels as characteristics. Following the division of the pictures into fixed sized meshes, the features are extracted. Reference [11] Ambarish Parida and Hima Bindu Maringanti's work "Hybrid Approach for Odia Handwritten Character Recognition" from

2014 suggests a hybrid technique for identifying handwritten Odia characters, integrating both structural and statistical aspects, in order to increase accuracy in character identification tasks. The strategy makes use of both approaches' strong points to improve Odia script recognition.

III. METHODOLOGY

A. Data preprocessing

The data set has 3 rows and 51 columns and it contains 48 unique label values. Then it goes through min max normalization. It is one of the most popular techniques for normalizing data. The minimum and maximum values of each characteristic are transformed to zero and one, respectively, while all other values are converted to a decimal between 0 and 1.

B. Feature Selection

It is a technique that reduces the input variable for your model by just using pertinent data and eliminating noise from the data. It involves selecting appropriate features for machine learning model automatically, depending on the kind of problem attempting to answer.

C. Classification

In the supervised machine learning process of classification, the model attempts to predict the correct label of a given input set of data. Prior to being utilized to make predictions on fresh, unobserved data, the model in classification is thoroughly trained utilizing the training set and then assessed using test data.

Logistic regression is known as statistical technique. Frequently used in binary classification problems to try and predict whether an instance belongs to a certain class or not. By evaluating the correlation between a collection of independent variables and a binary dependent variable, this approach yields a probabilistic result. The logistic function is used by the logistic regression model to transform a linear combination of input data into a probability score that is thresholded to give a binary result. Logistic regression is a useful tool for prediction and classification problems because it is straightforward to use, successful in situations where the result is binary, and easy to understand. It is frequently employed during the decision-making process.

Support Vector Machine (SVM) is a versatile machine learning technique that can handle both regression and classification issues. But its principal advantage and fundamental design is in how well it addresses classification problems. Because SVM can locate the optimal hyperplane in an N-dimensional space to split data points into distinct classes in feature space, it excels at classification. In an attempt to provide outstanding generalization to previously unidentified data, the method optimizes the margin between several classes. SVM works better in classification jobs because, while it can be customized for regression tasks as well, its main use is in scenarios where the objective is to discover patterns and boundaries between different categories within the data. The algorithm's

significance for applications needing precise categorization determinations is shown by its ability to determine the most discriminative hyperplane.

Decision trees may be used for regression tasks as well as the frequent supervised learning problem-solving application of them in classification. This classifier is represented as a tree structure, with each leaf node indicating a particular result, branches signifying decision rules based on aspects of the dataset, and core nodes representing qualities that are used to form conclusions. When it comes to predicting numerical results and classifying occurrences into discrete groups, decision trees excel in regression settings. Decision trees are especially useful for evaluating and elucidating the reasoning behind model predictions because of their hierarchical design, which makes decision logic evident. Decision trees are a commonly used tool in a wide range of industries due to their adaptability and interpretability, which provide insights into the interactions between various aspects.

Random Forest is a flexible and popular machine learning technique that combines the predictions of several decision trees to get a consistent and accurate result. Its versatility, simplicity of use, and capacity to handle a wide range of tasks, including regression and classification problems, are the main elements boosting its popularity. Random Forest reduces overfitting and improves generalization performance by building an ensemble of decision trees and aggregating their outputs using an average for regression or a voting method for classification. Its ease of use and resilience to noisy data further contribute to its wide range of applications across different domains, making it a top pick for practitioners looking for a reliable and efficient predictive modeling tool.

K-Nearest Neighbors (KNN) is a straightforward and user-friendly machine learning technique that can be used to both regression and classification issues. Predictions made using this non-parametric method are based on the majority class in the case of classification or the average value for regression, which is derived from the k-nearest data points in the feature space. The fundamental concept relies on closeness: each instance in the dataset is represented in a multi-dimensional feature space, and the forecast for each data point is based on the average or consensus of its k-nearest neighbors. KNN's adaptability, ease of use, and clear concept make it a popular choice for a variety of applications. It offers a useful method of making predictions without the need for explicit model training.

Multi-layer perceptron (MLP) kind of neural network consists of three layers of nodes: an input layer, one or more hidden layers, and an output layer. By allowing information to flow in a single direction from the input layer through the hidden layers and out to the output layer, MLPs operate as feedforward neural networks. To optimize the network's ability to learn and produce accurate predictions, weights are provided to the connected nodes, or neurons, that comprise each layer and are modified throughout the training phase. MLPs are helpful tools for a range of applications, such as pattern recognition, regression analysis, and classification, since they

are good at finding complex connections and patterns in data. The depth and connectivity of MLPs make them to acquire the ability to understand hierarchical data structures, which enables them to solve complex issues in a variety of fields.

Naive Bayes classifier is a probabilistic classification method based on feature independence and the Bayes theorem. Because it provides an easy-to-use and efficient approach for text classification, Naive Bayes is a highly helpful tool. It makes the assumption that the characteristics used for classification are conditionally independent in order to streamline the probability computation. The classifier determines the likelihood of a certain class given a set of attributes by combining prior probabilities and conditional probabilities using the Bayes theorem. Naive Bayes is a well-liked solution for problems like spam identification, sentiment analysis, and document classification in natural language processing applications, despite its "naive" assumption that features are independent. In real-world scenarios, it often performs exceptionally well.

Extreme Gradient Boosting, or XGBoost, is one of the most popular and successful machine learning methods. It is particularly good at handling issues with tabular and structured information fast. XGBoost is a member of the gradient boosting family. It constructs a set of weak learners, usually decision trees, then combines their predictions skillfully to produce an accurate and dependable prediction model. The method's careful integration of several weak learners, which enables it to continually repair faults and improve expected performance, makes it very successful in predictive modeling, classification, and regression tasks on structured data.

CatBoost, which stands for "Categorical Boosting," is an open-source, high-performance gradient boosting toolkit that was created specifically for machine learning applications. Its primary advantage is its exceptional capacity to handle categorical data efficiently and with minimum advance planning. CatBoost works incredibly well at producing reliable and accurate predictions when applied to datasets with categorical variables. The automatic categorical data processing of this library makes it possible to add features of that kind to the machine learning model without the need for human encoding or modification. Because it prioritizes performance and usability, CatBoost is a helpful tool for practitioners searching for reliable and efficient gradient boosting solutions. This is particularly true when the predictive modeling procedure makes use of categorical data.

Adaptive boosting, or AdaBoost for short, is an ensemble learning approach that combines the output of many decision trees, or weak learners, to create a robust and accurate model with the goal of increasing the accuracy of prediction models. In AdaBoost, each occurrence in the dataset is assigned a weight. During the training phase, the algorithm adjusts these weights to emphasize instances of inaccurate classifications. AdaBoost's adaptive feature allows it to focus on scenarios that were difficult for the previous weak learners, which allows it to progressively enhance the model's performance. To produce the final predictive model, the predictions made

by the weaker learners are put together and weighted, with larger weight going to those who make more accurate predictions. AdaBoost's power resides in its capacity to learn from errors iteratively, resulting in a robust and adaptable ensemble model that can handle challenging datasets and raise overall prediction accuracy.

D. Grid Search Cross-Validation

Grid Search Cross-Validation (Grid Search CV) is a technique used to find the optimal hyperparameters for a machine learning model by searching through a predefined set of hyperparameter values. It systematically evaluates the model's performance for each combination of hyperparameters, making it a powerful tool for hyperparameter tuning.

E. LIME(Local Interpretable Model-agnostic Explanations)

LIME (Local Interpretable Model-agnostic Explanations) is a technique used to explain the predictions of machine learning models, especially black-box models. It provides interpretable rationale for individual forecasts by locally approximating the complex model's decision boundary.

F. Proposed Methodology

The 'Malayalam Char Gabor' dataset is used in the proposed study to identify Malayalam palm leaves. To get the dataset ready for analysis, initial data preparation includes feature selection, min-max normalization, and null value elimination. Classifier models are compared with respect to accuracy, precision, recall, F1-score, and other parameters. Next, in order to improve model performance, hyperparameter adjustment is applied, and the classifiers are reassessed. After the best classifier has been chosen, Lime (Local Interpretable Model-agnostic Explanations) is used to give comprehensible explanations for each prediction the machine learning model makes. This all-encompassing strategy seeks to not only determine the best classifier for Malayalam palm leaf recognition but also to improve model interpretability using Lime, illuminating the variables impacting individual predictions in the context of the 'Malayalam Char Gabor' dataset.

IV. RESULTS AND ANALYSIS

The classifier models like Logistic Regression,Support Vector Machine(SVM),Random Forest Classifier,Decision Tree,K-Nearest Neighbour (KNN),Multi Layer Perceptron (MLP),Navie Bayes,XGBoost,CAT Boost,ADA Boost are compared with each other.

[Fig.01],shows the metrices and the classifier models which has been compared without Hyper parameter Tuning.Random Forest was observed to have the highest test accuracy (86.75%) while ADA Boost followed the accuracy of (85.95%).Random Forest had the highest F1 score(82.73%) while ADA Boost has second higher F1 score of (81.68%).

[Fig.02],shows the metrices and the classifier models which has been compared with Hyper parameter Tuning.Random Forest was observed to have the highest test accuracy (86.69%) followed by KNN and Decision Tree (Both) with a test

Classifiers	Report before Hyper parameter tuning							
	Accuracy		Precision		Recall		F1-Score	
	Test	Train	Test	Train	Test	Train	Test	Train
KNN	72.09%	81.25%	66.01%	76.16%	69.17%	77.78%	67.56%	76.96%
RF	86.75%	92.91%	82.62%	88.49%	82.84%	89.89%	82.73%	89.18%
Decision Tree	85.83%	92.91%	80.78%	88.49%	81.68%	89.96%	81.23%	89.22%
MLP	20.12%	21.08%	12.36%	13.00%	17.71%	18.25%	14.56%	15.19%
SVM	17.79%	19.18%	9.33%	10.74%	15.63%	16.68%	11.69%	13.07%
Naive Bayes	1.78%	2.64%	0.04%	0.05%	2.08%	2.08%	0.07%	0.11%
LR	10.31%	10.77%	2.51%	2.51%	9.09%	9.00%	3.93%	3.93%
XG-Boost	73.87%	83.98%	69.28%	80.10%	69.67%	80.23%	69.48%	80.16%
CAT-Boost	33.87%	37.72%	29.35%	33.91%	30.40%	33.30%	29.87%	33.60%
ADA Boost	85.95%	92.91%	81.43%	88.49%	81.93%	89.96%	81.68%	89.22%

Fig. 1. Without Hyper Parameter Tuning

Classifiers	Report after Hyper parameter tuning							
	Accuracy		Precision		Recall		F1-Score	
	Test	Train	Test	Train	Test	Train	Test	Train
KNN	86.26%	92.88%	81.81%	88.41%	82.41%	89.90%	82.11%	89.15%
RF	86.69%	92.91%	82.44%	88.45%	82.74%	89.93%	82.74%	89.18%
Decision Tree	86.26%	92.91%	81.23%	88.49%	82.02%	89.96%	81.62%	89.22%
MLP	19.02%	19.47%	7.26%	7.78%	16.15%	16.43%	9.01%	9.14%
SVM	22.64%	24.41%	22.11%	25.70%	20.55%	21.04%	20.55%	23.14%
Naive Bayes	16.87%	17.45%	13.50%	14.53%	15.84%	16.61%	14.58%	15.50%
LR	11.78%	11.69%	5.13%	5.32%	11.31%	11.13%	6.04%	5.98%
XG-Boost	75.03%	84.44%	70.97%	80.69%	71.04%	80.75%	71.00%	80.72%
CAT-Boost	55.34%	64.23%	47.19%	57.26%	52.03%	59.85%	49.49%	58.53%
ADA Boost	86.01%	92.91%	81.46%	88.49%	81.94%	89.96%	81.70%	89.22%

Fig. 2. With Hyper Parameter Tuning

accuracy of (86.26%).Random Forest was observed to have the highest F1Score (82.74%) followed by KNN F1Score(82.11%) and Decision Tree F1Score(81.62%)

[Fig.3]Lime is highly useful in identifying the significance of traits in predicting outcomes when probabilities are incorporated into the categorization process. Of all the features in the analysis, feature 9 has the highest probability contribution 49% according to Lime's feature significance rating. This suggests that it significantly affects how the model makes decisions. Feature 44 comes in second place with a chance of 36%. These insights provide a clear understanding of the relative value of different variables in influencing the model's predictions, as well as an accessible and intelligible look at the factors that substantially contribute to the biggest probability of specific outcomes. These feature-level explanations provide consumers a clearer picture of the internal workings of the system.

V. CONCLUSION

After examining a number of methods, it is clear that Random Forest outperforms the other algorithms in terms of accuracy, with K-Means trailing closely behind. Next, Lime, a tool for model interpretability, is used to characterize the elements influencing the model's predictions. Given that fea-

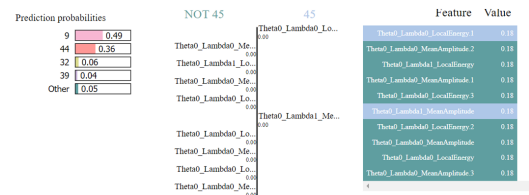


Fig. 3. Lime

ture 9 increases the chance to 49%, the evidence suggests that it is important. This important realization deepens our comprehension of the characteristics that significantly influence the model's output. After that, lime is used to intentionally decode palm leaf messages. Rather than using accuracy measures, its application depends on character recognition on palm fronds. With the use of this program, palm leaf health may be evaluated, characteristics associated with incompleteness, fracture, and brittleness can be identified, and potential contamination and erosion by microorganisms. Lime's interpretability facilitates in-depth research and provides insightful information on the nuances of palm leaf characteristics.

In essence, combining Lime's application, feature interpretation, and algorithmic analysis results in a comprehensive comprehension of the model's functioning and its suitability for defining the state of palm leaves. A comprehensive plan can help with issues like erosion and brittleness, enabling focused treatments and educated decisions on palm leaf research and preservation.

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